

CUSTOMER CHURN ANALYSIS

Project Report

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Tools Used:

Python | SQLite | Pandas | NumPy | Matplotlib | Seaborn

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Project Overview

Customer churn is a major challenge for subscription-based businesses such as OTT platforms, SaaS companies, telecom providers, and streaming services. This project analyzes customer data to identify churn patterns, measure customer retention, and generate business insights that help improve customer retention strategies.

The project uses data stored in a SQLite database, which is imported into Python for data cleaning, feature engineering, exploratory data analysis, and visualization.

Business Objective

The objectives of this project are to:

- Identify customers who have churned.
 - Measure overall churn and retention rates.
 - Analyze customer behavior across different subscription plans and regions.
 - Identify high-risk customers.
 - Provide actionable recommendations to improve customer retention.
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Tools & Technologies

- Python
 - SQLite (Database)
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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Dataset

The project uses a SQLite database (**customer_churn.db**) containing three related tables.

Customer Table

- Customer ID
- Customer Name
- Country
- State
- Gender
- Date of Birth

Subscription Table

- Subscription Start Date
- Subscription Type
- Plan Type
- Contract Type
- Monthly Charges
- Customer Lifetime Value (CLTV)
- Churn Score
- Cancellation Details

Support Table

- Complaint Date
 - Escalation Status
 - Customer Satisfaction (CSAT)
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Project Workflow

1. Data Import

- Connected Python to the SQLite database using the `sqlite3` library.
- Imported customer, subscription, and support tables into Pandas DataFrames.
- Combined the tables using customer ID for further analysis.

2. Data Cleaning

Performed several data cleaning tasks, including:

- Renamed columns for better readability.
- Removed unnecessary columns.
- Converted date columns into datetime format.
- Standardized gender values.
- Filled missing country values using existing state information.
- Removed duplicate support records before merging datasets.

3. Feature Engineering

Created new business features such as:

- Churn Flag
- Complaint Count
- Customer Tenure
- Churn Risk Category

These features helped improve the analysis and customer segmentation.

4. Data Analysis

Calculated important business KPIs, including:

- Churn Rate
- Retention Rate
- Churn by Plan Type
- Churn by State
- Revenue at Risk
- Average Revenue Per User (ARPU)
- Average Customer Tenure
- Escalation Rate
- Average Complaints per Customer
- Correlation between Escalations and Churn

5. Data Visualization

Created visualizations to better understand customer behavior, including:

- Monthly Churn Trend
- Churn Rate by Plan Type
- Churn by State
- Correlation Heatmap
- Pair Plot
- Category-wise Analysis

Key Findings

- Overall Churn Rate: **28.57%**
- Retention Rate: **71.43%**
- Basic subscription plans showed the highest churn rate.
- Karnataka recorded the highest customer churn.
- Monthly contract customers churned more frequently than annual contract customers.
- Average Revenue Per User (ARPU): **18.85**
- Average Customer Tenure: **Approximately 1,490 days**
- Revenue at Risk due to churn: **73.94**
- Customers with support escalations showed a strong positive relationship with churn.

Business Recommendations

- Improve customer experience for Basic plan subscribers.
- Investigate the reasons for high churn in Karnataka.
- Encourage customers to choose annual subscription plans.
- Prioritize customers with Medium and High churn risk for retention campaigns.

- Resolve customer complaints quickly to reduce churn.
 - Monitor customer satisfaction and escalation trends regularly.
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Project Outcome

This project demonstrates how customer data stored in a SQLite database can be extracted into Python, cleaned, transformed, and analyzed to generate meaningful business insights. The analysis helps identify churn drivers, estimate revenue loss, and support data-driven decisions for improving customer retention.

Skills Demonstrated

- Python Programming
 - SQLite Database Connectivity
 - Data Cleaning
 - Data Transformation
 - Feature Engineering
 - Exploratory Data Analysis (EDA)
 - KPI Development
 - Data Visualization
 - Customer Analytics
 - Business Intelligence
 - Analytical Thinking
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Conclusion

This project presents an end-to-end customer churn analysis workflow using Python. By importing data from a SQLite database, performing data preprocessing, feature engineering, KPI analysis, and visualization, the project provides valuable insights into customer behavior and churn patterns. The findings can help businesses make informed decisions to improve customer retention and reduce revenue loss.